



Chemical Theory II

Statistical

Thermodynamics

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The course is developed based on the
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Summary of last lecture

- Interpretation of q
 - $q' \approx$ number of thermally accessible states
- Replacing the partition function sum with an integral
 - Fine if $q' \gg 1$.
- Translational partition function
 - Particle in a three-dimensional box.
 - $q_{\text{trans}} = V / \Lambda^3$
 - Typical value 10^{30} .
- Rotational partition function
 - Rigid rotor: $(2J + 1)$ -fold degenerate levels.
 - The “rotational temperature” $\theta_{\text{rot}} = B / k$
 - $q_{\text{rot}} = T / \sigma \theta_{\text{rot}}$
 - Much smaller than q_{trans} but usually $q_{\text{rot}} \gg 1$
 - Explicit sum may be necessary at low T or large B .

Rotational partition function for diatomics

- So that
$$q_{\text{rot}} = \left[\frac{-kT}{B} \exp\left(\frac{-BJ(J+1)}{kT}\right) \right]_0^\infty$$
$$= \frac{kT}{B}.$$
- It is common to define *a rotational temperature, θ_{rot}* , as
$$\theta_{\text{rot}} = \frac{B}{k};$$

the name derives from the fact that θ_{rot} *has the dimension of temperature*. The rotational partition function can thus be written:

$$q_{\text{rot}} = \frac{kT}{\sigma B} \quad \text{or} \quad q_{\text{rot}} = \frac{T}{\sigma \theta_{\text{rot}}}.$$

Where the *symmetry parameter, σ* , has been introduced. *It takes the value 2 for homonuclear diatomics and 1 for heteronuclear diatomics.*

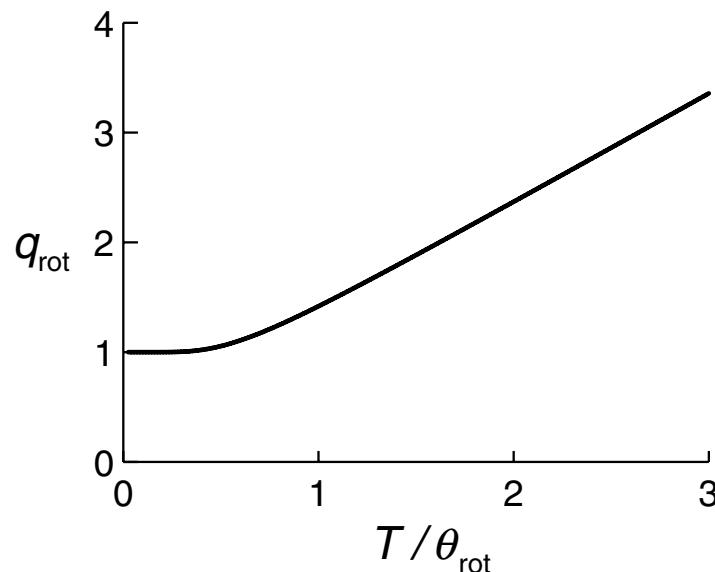
- For CO, with $B \sim 1.925 \text{ cm}^{-1}$, the rotational temperature is easily calculated if we use the value of k in $\text{cm}^{-1} \text{ K}^{-1}$ ($k = 0.6950 \text{ cm}^{-1} \text{ K}^{-1}$). Hence $\theta_{\text{rot}} = 1.925/0.695 = 2.77 \text{ K}$ and $q_{\text{rot}} = 108$ at 298 K. *The rotational partition function is very much smaller than the translational partition function; nevertheless the fact that many rotational states contribute means that q_{rot} is significantly greater than 1.*

Example

- The rotational constant of I_2 is 0.0373 cm^{-1} ; compute the rotational partition function at 298 K.

Rotational partition function for diatomics

- *If it is not the case that $T > \theta_{\text{rot}}$, the equation derived from integration cannot be used and the rotational partition function has to be evaluated by computing successive terms in the sum until they are insignificant. In practice only a few terms are needed as in this limit only a few levels are occupied. Shown below is a plot of q_{rot} as a function of T / θ_{rot} ; as expected the partition function is 1 when $T / \theta_{\text{rot}} \ll 1$. The value of q_{rot} rises quickly once $T / \theta_{\text{rot}} \approx 1$ and at higher temperatures the partition function rises linearly with T as predicted by the formula from integration.*
- The rotational energy levels of linear polyatomic molecules are identical to those for a diatomic; however the expression for the moment of inertia is more complex.



Rotational partition function for non-linear molecules

- Non-linear molecules have three moment of inertia, denoted I_a , I_b and I_c . Associated with each is a rotational constant, for example

$$\tilde{A} = \frac{h}{8\pi^2 \tilde{c} I_a} \quad \tilde{c} \text{ in cm s}^{-1}, \quad I_a \text{ in kg m}^2$$

- In turn each of the rotational constants is associated with a rotational temperature, for example

$$\theta_{\text{rot},a} = \frac{\tilde{A}}{k}.$$

- In the high temperature limit where $T \gg \theta_{\text{rot}}$ it can be shown that the rotational partition function is given by

$$q_{\text{rot}} = \frac{\pi^{\frac{1}{2}}}{\sigma} \left(\frac{T^3}{\theta_{\text{rot},a} \theta_{\text{rot},b} \theta_{\text{rot},c}} \right)^{\frac{1}{2}}$$

where σ is the symmetry number. *For H_2O σ is two, and for NH_3 σ is three.*

Vibrational partition function

- The model we use for the energy levels of a vibrating bond is the simple harmonic oscillator (SHO). This has energies E_v

$$E_v = \left(v + \frac{1}{2}\right) \hbar \omega$$

- where v is the vibrational quantum number, taking values 0, 1, 2 . . . and ω is the vibrational frequency. ω is in angular frequency units (rad s^{-1}), and is therefore related to the frequency of the vibration in Hz, ν_0 , by $\omega = 2\pi\nu_0$. So, the energy levels can be written

$$E_v = \left(v + \frac{1}{2}\right) h\nu_0$$

- These energies are measured relative to the bottom of the potential energy well, that is the minimum in the potential when the bond is at its equilibrium length. *The energy of the ground state, $v = 0$, is $1/2 h\nu_0$.* If we measure the energies relative to this state (i.e. define the energy of the ground state to be zero) the energy levels are

$$E_v = v h\nu_0 \quad [\text{relative to ground state}]$$

Vibrational partition function

- Using these levels the partition function is

$$q'_{\text{vib}} = \sum_{v=0}^{\infty} \exp\left(\frac{-v h \nu_0}{kT}\right)$$

and the first few terms are

$$q'_{\text{vib}} = 1 + \exp\left(\frac{-h \nu_0}{kT}\right) + \exp\left(\frac{-2h \nu_0}{kT}\right) + \dots$$

- As before, we first consider the spacing of the energy levels compared to kT ; the spacing is simply $h \nu_0$. It is more convenient to express the energies in cm^{-1} , in which case $E_v = (v + 1/2) \tilde{\omega}$, where $\tilde{\omega}$ is the vibrational frequency in cm^{-1} ; the spacing is therefore $\tilde{\omega}$.
- *Typical diatomics have vibrational frequencies in the range 200 – 4000 cm^{-1} . For all but the lowest vibrational frequencies the spacing is much greater than kT at room temperature ($\approx 200 \text{ cm}^{-1}$), so only one or two terms contribute to the partition function. There is no question of replacing the sum by an integral.*

Vibrational partition function

- However, it turns out that the sum for q_{vib} can be evaluated in a closed form by recognizing that the terms in the equation above form a geometric progression. A *geometric progression (GP)* is a series in which successive terms are found by multiplying the previous term by r , the common ratio; if the first term is a , then the GP is

$$a + ar + ar^2 + ar^3 + \dots$$

- If the common ratio is less than one, $r < 1$, successive terms get smaller and smaller; under these circumstances the sum of an infinite number of terms is finite and given by

$$\frac{a}{1 - r}.$$

Vibrational partition function

- The sum for q'_{vib} is a GP with $a = 1$ and $r = \exp(-h\nu_0/kT)$. The common ratio is < 1 , regardless of the value of ν_0 or T , so the sum to infinity of the series is given by

$$q'_{\text{vib}} = \frac{1}{1 - \exp\left(\frac{-h\nu_0}{kT}\right)}.$$

- It is common to define a vibrational temperature, θ_{vib} , as

$$\theta_{\text{vib}} = \frac{h\nu_0}{k}$$

- or, if the vibrational frequency is expressed in wavenumbers, $\theta_{\text{vib}} = \tilde{\omega} / k$ where $k = 0.6950 \text{ cm}^{-1} \text{ K}^{-1}$. Using this, the partition function can be written as

$$q'_{\text{vib}} = \frac{1}{1 - \exp\left(\frac{-\theta_{\text{vib}}}{T}\right)}.$$

Vibrational partition function

- To find the partition function for the case where the energies of the levels are measured from the bottom of the potential energy well, we simply obtain (NB $\epsilon^0 = 1/2 h\nu_0$):

$$q = q' \exp\left(\frac{-\epsilon^0}{kT}\right)$$

- So, the partition function using energy levels measured from the bottom of the potential well is

$$q_{\text{vib}} = \frac{\exp\left(\frac{-h\nu_0}{2kT}\right)}{1 - \exp\left(\frac{-h\nu_0}{kT}\right)} = \frac{\exp\left(\frac{-\theta_{\text{vib}}}{2T}\right)}{1 - \exp\left(\frac{-\theta_{\text{vib}}}{T}\right)}.$$

- For CO, which has a vibrational frequency of 2157 cm^{-1} , $\theta_{\text{vib}} = 2157/0.6950 = 3079 \text{ K}$. With such a high vibrational temperature, at 298 K only the first term contributes to q_{vib} and hence $q'_{\text{vib}} = 1.000$. For I_2 , the vibrational frequency is 130 cm^{-1} , $\theta_{\text{vib}} = 130/0.6950 = 187 \text{ K}$. At 298 K , θ_{vib} is comparable to the temperature and so q'_{vib} is expected to exceed 1; the value is 2.15.

Example

- The vibrational frequency of HCl is 2938 cm^{-1} . Compute the vibrational partition function at 298 K for the case (a) where the ground state is taken as the energy zero and (b) where the energy zero is taken as the bottom of the potential energy well.

More complex molecules

- The vibrations of more complex molecules can be described using normal modes. A normal mode is a motion of the atoms in a molecule in which
 - all the atoms move at the same frequency
 - the centre of mass remains fixed in space
 - the motions are simple harmonic
- *For a molecule with N atoms there are $(3N - 6)$ normal modes, except in the case of a linear molecule when there are $(3N - 5)$.*
- Each normal mode has associated with it a set of energy levels which can be modelled using a harmonic oscillator; the vibrational frequency for each normal mode is different (except in the case of degeneracy)

More complex molecules

- The vibrational partition function for the i th normal mode, $q_{\text{vib},i}$, is as derived just now. For the case where the energies are measured relative to the bottom of the potential well, we have

$$q_{\text{vib},i} = \frac{\exp\left(\frac{-\theta_{\text{vib},i}}{2T}\right)}{1 - \exp\left(\frac{-\theta_{\text{vib},i}}{T}\right)}$$

- where $\theta_{\text{vib},i}$ is the vibrational temperature for the i th normal mode with frequency $\nu_{0,i}$ ($\theta_{\text{vib},i} = h\nu_{0,i} / k$ or $\theta_{\text{vib},i} = \tilde{\omega}_i / k$). *As the energy levels for each normal mode are separate, the total vibration partition function is found by multiplying the $q_{\text{vib},i}$ together for each normal mode:*

$$q_{\text{vib}} = q_{\text{vib},1} q_{\text{vib},2} q_{\text{vib},3} \dots$$

- If a particular mode is n -fold degenerate it contributes n identical terms to this product. For example, in the case of CO_2

$$q_{\text{vib}} = q_{\text{vib},\text{sym.str.}} q_{\text{vib},\text{asym.str.}} q_{\text{vib},\text{bend}} q_{\text{vib},\text{bend}} = q_{\text{vib},\text{sym.str.}} q_{\text{vib},\text{asym.str.}} (q_{\text{vib},\text{bend}})^2.$$

Electronic partition function

- Except in the case of hydrogen atoms, there is no simple quantum mechanical model that can be used to determine the energies of electronic levels. So, these have to be determined by experiment (spectroscopy). It is important to remember that electronic states may be degenerate so that in evaluating q_{elec} is appropriate

$$q_{elec} = \sum_j g_j \exp\left(\frac{-\epsilon_{elec,j}}{kT}\right)$$

- If the energies of the electronic states are measured relative to the ground state (i.e. $\epsilon_{elec,0} = 0$), then the electronic partition function becomes

$$q'_{elec} = g_0 + g_1 \exp\left(\frac{-\epsilon_{elec,1}}{kT}\right) + g_2 \exp\left(\frac{-\epsilon_{elec,2}}{kT}\right) + \dots$$

- It is often the case that the excited electronic levels lie at energies much greater than kT above the ground state. In such cases only the first term contributes and $q'_{elec} = g_0$, the degeneracy of the ground state.
- Of course, the electronic ground state does not actually have energy zero, and different atoms and molecules will have different ground state energies. It is important to take this into account when considering chemical equilibrium.

Atoms

- The electronic states of atoms are characterised by term symbols, which give the values of ***S***, ***L*** and ***J***, *the total spin angular momentum, the total orbital angular momentum and the total angular momentum*, respectively: $^{2S+1}L_J$.
- The degeneracy of such a level is **$(2J + 1)$** . For example, Na atoms have a ground state with term symbol $^2S_{1/2}$; thus $J = 1/2$ and the degeneracy is $2 \times 1/2 + 1 = 2$ so $q'_{elec} = 2$ (the excited states are too high in energy to contribute).

Molecules

- Most molecules have closed shell electronic configurations (i.e. all the occupied orbitals have pairs of electrons in them, and any group of degenerate orbitals is fully occupied). The resulting electronic states are not degenerate and so $q'_{elec} = 1$.
- An exception is O_2 , whose ground state term symbol is $^3\Sigma^-_g$; the degeneracy of this state is 3, so $q'_{elec} = 3$.

Isotopic substitution

- Changing an atom in a molecule from one isotope to another (for example H to D or ^{12}C to ^{13}C) does not alter the number of electrons or the nuclear charges. We therefore expect that, to a very good approximation, the bond lengths and force constants will remain the same.
- As the rotational constant is proportional to $1/I$ and $I = \mu R^2$, it follows that, for an isotopic substitution,

$$\frac{B_A}{B_B} = \frac{\mu_B}{\mu_A}$$

where the subscripts A and B refer to different isotopic species.

- The vibrational frequency depends on $(k_{\text{force}}/\mu)^{1/2}$, where k_{force} is the force constant. If it is assumed that the force constant remains the same with isotopic substitution then

$$\frac{\nu_{0,A}}{\nu_{0,B}} = \frac{\tilde{\omega}_A}{\tilde{\omega}_B} = \sqrt{\frac{\mu_B}{\mu_A}}.$$

Course outline

- Separation contributions to thermodynamic functions

Separation contributions to thermodynamic functions

- We saw previously that the molecular partition function could be written as a product of contributions from different molecular energy levels

$$q = q_{\text{trans}} q_{\text{rot}} q_{\text{vib}} q_{\text{elec}}$$

- and we also saw that thermodynamic functions can be expressed in terms of q (for indistinguishable particles)

$$A = -kT \ln Q_N = -kT \ln \frac{q^N}{N!}$$

- As q is a product of terms associated with different kinds of energy, it follows that A will be a sum of terms (due to the logarithm)

$$A = A_{\text{trans}} + A_{\text{rot}} + A_{\text{vib}} + A_{\text{elec}}.$$

Separation contributions to thermodynamic functions

- *There is only one factor of $N!$ and it is usual to associate this with q_{trans} , so that*

$$\begin{aligned} A_{\text{trans}} &= -kT \ln \frac{q_{\text{trans}}^N}{N!} \\ &= -NkT \ln q_{\text{trans}} + kT (N \ln N - N) \end{aligned}$$

- The other terms take the form

$$\begin{aligned} A_{\text{rot}} &= -kT \ln q_{\text{rot}}^N \\ &= -NkT \ln q_{\text{rot}} \end{aligned}$$

and likewise for the vibrational and electronic contributions.

Separation contributions to thermodynamic functions

- A similar line of argument is used for the contributions to the entropy:

$$\begin{aligned} S_{\text{trans}} &= k \ln \frac{q_{\text{trans}}^N}{N!} + kT \left(\frac{\partial \ln q_{\text{trans}}^N / N!}{\partial T} \right)_V \\ &= Nk \ln q_{\text{trans}} - Nk \ln N + Nk + NkT \left(\frac{\partial \ln q_{\text{trans}}}{\partial T} \right)_V \end{aligned}$$

$$\begin{aligned} S_{\text{rot}} &= k \ln q_{\text{rot}}^N + kT \left(\frac{\partial \ln q_{\text{rot}}^N}{\partial T} \right)_V \\ &= Nk \ln q_{\text{rot}} + NkT \left(\frac{\partial \ln q_{\text{rot}}}{\partial T} \right)_V . \end{aligned}$$

Separation contributions to thermodynamic functions

- For *the case of internal energy the $N!$ term disappears* upon differentiation

$$U = kT^2 \left(\frac{\partial \ln q^N / N!}{\partial T} \right)_V = NkT^2 \left(\frac{\partial \ln q}{\partial T} \right)_V$$

- so for each type of energy level the expression for U is the same, for example

$$U_{\text{trans}} = NkT^2 \left(\frac{\partial \ln q_{\text{trans}}}{\partial T} \right)_V$$

and so on for the other types of energy levels.

Course outline

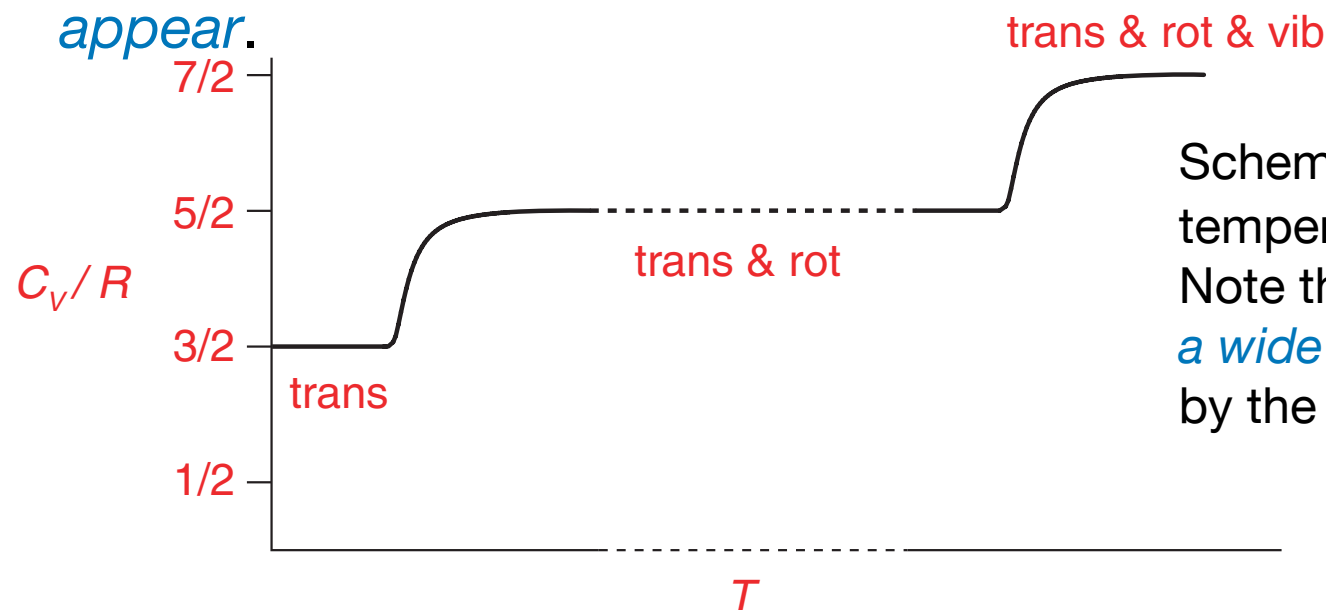
- Internal energy and heat capacities

Internal energy and heat capacities

- In this section we will look at the contributions that different kinds of energy levels make to the internal energy, U , and the (constant volume) heat capacity, C_V . This will be the first practical application of the relationships derived in the preceding lectures, and it will also give important insight into the relationship between partition functions and thermodynamic variables.

Internal energy and heat capacities

- One of the observations which we will be able to explain is the rather characteristic way in which *heat capacities of gases vary with temperature*. Typically, what is found is that *as the temperature is increased the heat capacity changes little until a particular point is reached at which the heat capacity quickly rises to a new value; after this it once again remains constant as the temperature increases*. At still higher temperatures a further step occurs. The result is *a series of plateaus*, illustrated below for a diatomic molecule. What is more remarkable is that *the plateau values are the same for all diatomic molecules; all that changes is the temperatures at which the steps appear*.



Schematic variation of C_v with temperature for a diatomic molecule. Note that *the central plateau covers a wide temperature range*, indicated by the dashed line.

Internal energy and heat capacities

- The interpretation of this behaviour is that *at the lowest temperatures only translational energy levels are occupied significantly and so only these contribute to the heat capacity. Eventually the temperature becomes high enough for rotational levels to be occupied and as a result the heat capacity increases. Finally, at sufficiently high temperatures vibrational energy levels are occupied and the heat capacity rises once more.*
- By computing the heat capacity in terms of the partition function we will be able to predict the form of this curve, explain why there are a series of plateaus, find out the value of the heat capacity at the plateaus and identify the temperatures at which the steps occur. This discussion will lead us to an understanding of the *equipartition principle* and how it can be applied.

Equipartition principle

- The *equipartition principle* states that *each ‘squared term’ contributes $1/2 RT$ (per mole) to the internal energy of the system*; **R** is the gas constant, in SI given in units of $\text{J K}^{-1} \text{mol}^{-1}$. In molecular units the contribution is *$1/2 kT$ per particle*.
- What is a ‘*squared term*’? *It is any term in the expression for the energy of the molecule or atom which depends on the square of a position or velocity variable.*
- The translational energy depends on the square of the velocity, and so qualifies as a squared term. *Motion in the x , y , and z directions is independent, so translation in three-dimensions gives three squared terms and thus an internal energy of $3/2 RT$. This is often expressed by saying that there are three translational degrees of freedom.*

Equipartition principle

- The rotational energy depends on the square of the angular velocity and so qualifies as a squared term. In general a molecule can rotate about three orthogonal axes; each rotation gives rise to a contribution of $1/2 RT$. *There are thus three rotational degrees of freedom, contributing a total of $3/2 RT$. A diatomic (or any linear molecule) is a special case as there is no rotation about the internuclear axis, so there are only two rotational degrees of freedom. The internal energy is thus RT .*
- A vibration has two squared terms: one for the kinetic energy, which goes as the square of the velocity, and one for the potential energy, which goes as the square of the displacement. *The vibration of a diatomic molecule thus contributes RT to the energy. In general, each normal mode contributes RT to the energy.*

Heat capacities

- If the translational contribution to U is $\frac{3}{2} RT$, the corresponding contribution to the heat capacity is

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V = \frac{d \frac{3}{2} RT}{dT} = \frac{3}{2} R.$$

Likewise, *for a diatomic, the rotational contribution is R and the vibrational contribution is also R .*

- We can now explain the form of the plot on the previous page. At the lowest temperatures only *translational levels* contribute and so the heat capacity is $\frac{3}{2} R$; this rises to $\frac{5}{2} R$ once the temperature is high enough for *the rotational levels* to be occupied and so contributing a further R to the heat capacity. At still higher temperatures, *vibrational levels* begin to be occupied, ultimately contributing a further R to the heat capacity and bringing the final value to $\frac{7}{2} R$.